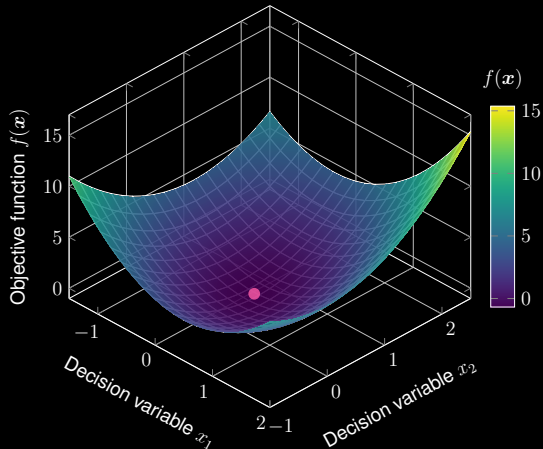
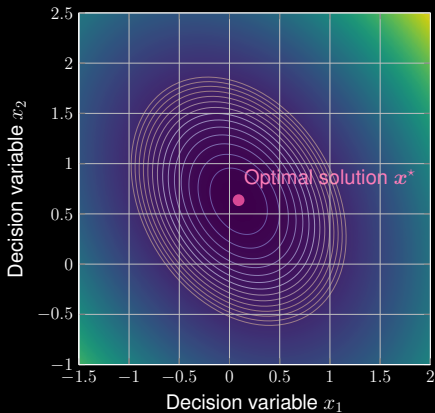


Quadratic Optimization

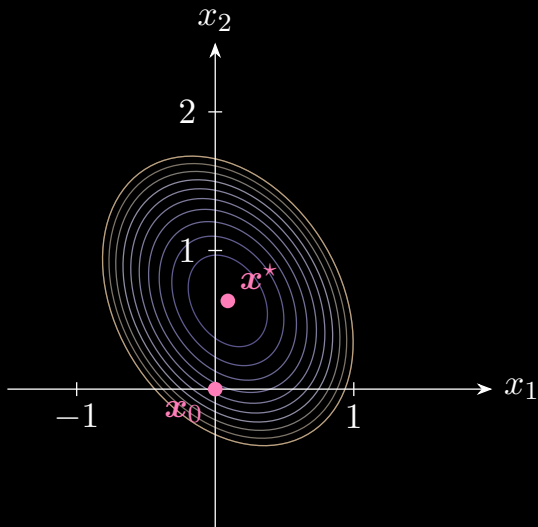
$$\min_{\mathbf{x} \in \mathbb{R}^2} \frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x} \quad \text{with} \quad \mathbf{A} = \begin{bmatrix} 4 & 1 \\ 1 & 3 \end{bmatrix} \quad \text{and} \quad \mathbf{b} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$



Quadratic Optimization

$$\min_{\mathbf{x} \in \mathbb{R}^2} \underbrace{\frac{1}{2} \mathbf{x}^\top \mathbf{A} \mathbf{x} - \mathbf{b}^\top \mathbf{x}}_{\text{Objective function } f} \quad \Rightarrow \quad \underbrace{\frac{\mathrm{d}f}{\mathrm{d}\mathbf{x}} = \mathbf{A} \mathbf{x} - \mathbf{b}}_{\text{First-order derivative}} = \mathbf{0} \quad \Rightarrow \quad \underbrace{\mathbf{A} \mathbf{x} = \mathbf{b}}_{\text{Linear system}}$$

Quadratic Optimization $\min_{x \in \mathbb{R}^2} \frac{1}{2} x^\top A x - b^\top x$



Thanks for your attention!

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